

Humza Jameel

AI Engineer | Full Stack Developer

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PROFILE

M.Sc. Artificial Intelligence student with 3+ years building full-stack applications and distributed AI workflows. Proven track record developing responsive React/TypeScript frontends for RAG pipelines and voice agents alongside high-performance Python/FastAPI backends on AWS. Skilled in agentic systems, event-driven architectures, and agile delivery with a first-principles approach.

TECHNICAL SKILLS

- **AI & Machine Learning:** LangChain, LangGraph, RAG, Multi-Agent Systems, VAPI.ai, PyTorch, Vector Search, Scikit-Learn, XGBoost, MLflow.
- **Backend & CS:** Python, FastAPI, Node.js, C# (.NET), AsyncIO, REST/GraphQL APIs, Structured Logging, Distributed Systems.
- **Frontend Development:** React, TypeScript, JavaScript (ES6+), Tailwind CSS, Next.js, Streamlit, Responsive UI Components.
- **Cloud, DevOps & Databases:** AWS (Lambda, S3, DynamoDB), Docker, CI/CD (GitHub Actions), Redis, PostgreSQL, MongoDB, Azure.

PROJECTS

DB Tight-Transfer Risk Predictor

XGBoost, FastAPI, MLflow, Docker, React, Pandas, scikit-learn

- Architected an end-to-end ML system predicting missed-connection risk across 12 Berlin rail hubs, processing 8.5M+ historical records into 175K+ labeled training events.
- Achieved 0.49 F1-score (65% recall) through MLflow-tracked XGBoost experimentation and deployed the model as a production FastAPI + Docker service with React frontend.

Agentic Voice Assistant with Tool-Calling

VAPI.ai, FastAPI, LangGraph, Python, AsyncIO

- Built a real-time voice assistant using async Python and LangGraph to orchestrate complex multi-step tool-calling workflows.
- Integrated external webhooks to synchronize live conversation states with ticketing management APIs.

MidiNet GAN — Symbolic Music Generation

PyTorch, GANs, MIDI, Python

- Reimplemented and optimized MidiNet GAN in PyTorch with chord conditioning; improved generation quality through GPU-based training experiments.

EXPERIENCE

AI / Full Stack Engineer (Working Student) | New Digital Intelligence GmbH, Berlin

May 2024 – Aug 2025

- Developed internal React/Node.js tooling to validate multi-step agentic behaviors and RAG features prior to production rollout.
- Built responsive interfaces for RAG pipelines and voice agents; implemented event-driven backend tasks with structured logging for observability.
- Evaluated emerging AI tools and frameworks to accelerate feature delivery within an agile team.

Product Development & Full Stack Engineer | Habib Bank Limited, Pakistan

Aug 2022 – Mar 2023

- Designed and maintained React features for large-scale banking web and mobile applications; integrated Redis caching to significantly reduce query latency.
- Refactored legacy code into reusable UI components and automated cloud CI/CD pipelines, accelerating release cycles and improving system reliability.

Software Engineer | United Bank Limited, Pakistan

Jul 2021 – Jan 2022

- Modernized financial web portals using JavaScript, HTML5 and CSS3; engineered reliable REST APIs with C# (.NET) and optimized database interactions.

Software Developer | Evantagesoft, Pakistan

Feb 2021 – Jul 2021

- Developed responsive MERN stack web applications with focus on component state management and clean API design.

EDUCATION

M.Sc. Artificial Intelligence

Apr 2023 – Present

Brandenburg University of Technology, Cottbus, Germany

Modules: Distributed Agentic Systems, Machine Learning, Automated Reasoning

B.Sc. Computer Science

Feb 2017 – Jan 2021

COMSATS University Islamabad, Pakistan

Thesis: Geo-navigation application (React Native + Node.js + MongoDB)

LANGUAGES

English: Fluent (C1) • German: A2 (Actively targeting B1)